

PAPER_883: Negative E(t) Buoyancy Erosion Master

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Abstract

Master equation for negative E(t) buoyancy-driven erosion. $E^-(t) = -E_0 \cdot \exp(\kappa t + [SSq] \cdot t/26) \cdot S_{26} \cdot (1 - F_{\{U,Bi\}}/F_U)$. Symmetric counterpart to $E^+(t)$, active when gravity dominates buoyancy (ratio < 0.5). Applies to filament erosion, photoevaporation in M16, and GW damping scenarios.

1. Core Equations

$E^-(t) = -E_0 \cdot \exp(\kappa t + [SSq] \cdot t/26) \cdot S_{26} \cdot (1 - R)$
net_factor = 2R - 1

2. Parameters

Parameter	Default	Description
E_0	1.0 J	Initial energy scale
t	0.0 s	Time parameter
F_UBi_over_FU	0.3	Buoyancy-to-field ratio

3. UQFF Integration

This calculator integrates negative_et_erosion.py from Session 205 into the CP4 pipeline as class #467.

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